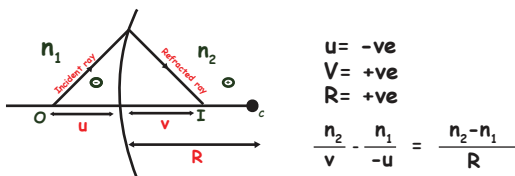


REFRACTION AT CURVED SURFACES



all lengths on the side of incident ray are taken as -ve
all lengths on the side of reflected ray are taken as +ve

$$\frac{I-O=S}{\text{Medium}} - \frac{\text{Medium}}{O\text{-Distance}} = \frac{\text{Change in medium}}{\text{Radius of curvature}}$$

Transverse Magnification $T.M = m = \frac{v}{u} = \frac{n_2 \times n_1}{u \times n_2}$

LENS MAKERS FORMULA

To find focal length if refractive index is same on both sides of lens.

$$I-O=S \quad \frac{n_1}{F} - 0 = \frac{n_2-n_1}{R_1} + \frac{n_1-n_2}{R_2}$$

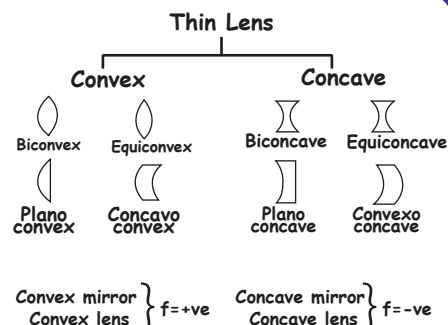
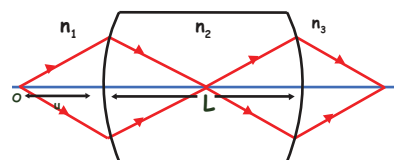
$$\frac{1}{F} = \left[\frac{n_2}{n_1} - 1 \right] \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

$$\frac{1}{F} = \left[n_2 - 1 \right] \left[\frac{1}{R_1} - \frac{1}{R_2} \right]$$

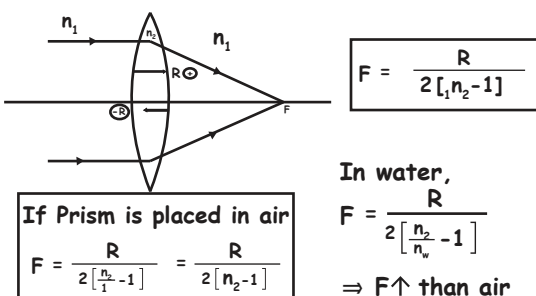
LENSES

Types

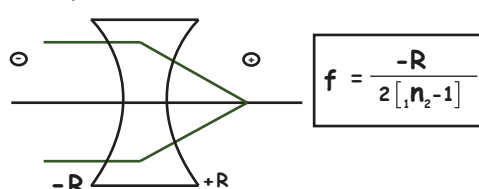
Thick lens - Two surfaces are at some distance apart.
Thin lens - Two surfaces are close



EQUICONVEX LENS



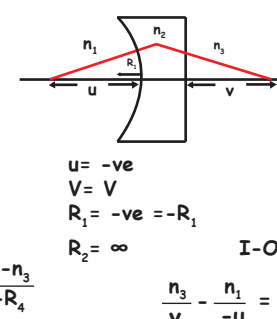
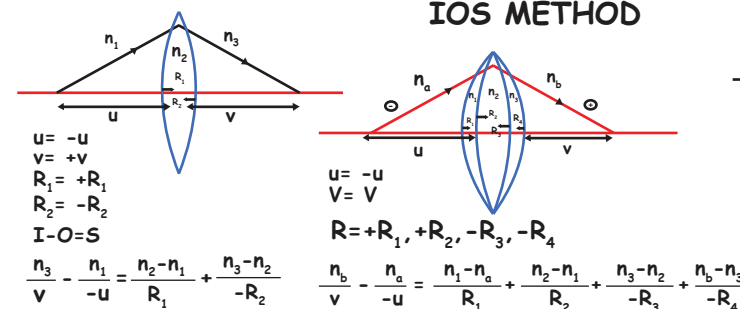
FOCAL LENGTH OF EQUICONCAVE LENS IN AIR



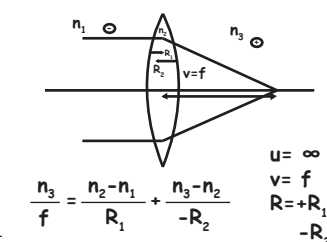
PLANO CONVEX LENS

F of planoconvex lens = 2 x f of equiconvex lens
 $f = \frac{R}{n_2 - 1}$

IOS METHOD



IOS METHOD FOR FOCAL LENGTH



PLANO CONCAVE LENS

Equiconvex = f → Equi Concave = -f
Plano Convex = 2f Plano Concave = -2f
 $f = -\frac{R}{n_2 - 1}$

POWER

Power = $\frac{1}{F \text{ in metre}}$ [Power = 1 dioptre = 1m⁻¹]
In centimetre, $P = \frac{100}{f \text{ in cm}}$

CUTTING OF LENS

	Before cutting	After cutting
Focal Length	f	2f
Power	P	P/2
Area	A	A
Intensity	I	I

	Before cutting	After cutting
Focal Length	f	f
Power	P	P
Area	A	A/2
Intensity	I	I/2

BLACKENING OF LENS



To find new intensity :

- Find total A_T (Area of lens before blackening)
- Find new A_T = (A_{total} - A_{opaque})
- I ∝ original A_T (Total A_T)

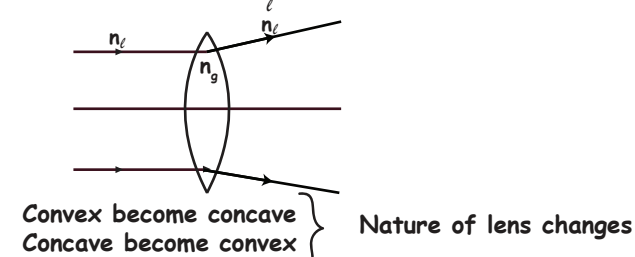
I' ∝ new A_T

Taking ratio of these two equations we can find I'

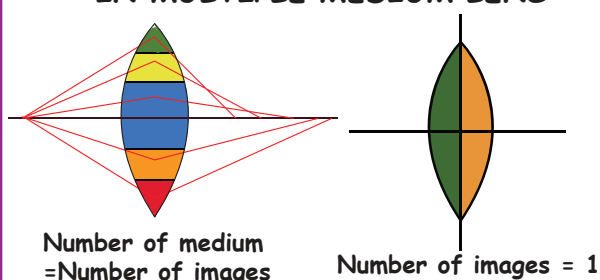
Comparison of focal length in air & Liquid

- If n_{air} < n_{liquid} < n_{glass}
 - Nature of lens remains same
 - Focal length increases
- Same refractive index [n₁ = n_{glass}]
 - Lens become invisible

3) n_l > n_g $f = \frac{R}{2(\frac{n_g}{n_l} - 1)}$ = negative



NUMBER OF IMAGES FORMED IN MULTIPLE MEDIUM LENS



MAGNIFICATION

To find size of image

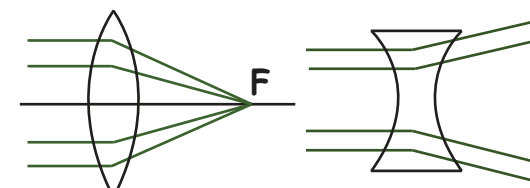
$$m = \frac{[I]}{[O]} = \frac{v}{u}$$

$$m = \frac{f}{f+u}$$

$$m = \frac{f-v}{f}$$

|m_T| = 1 ⇒ Same size
|m_T| < 1 ⇒ Small
|m_T| > 1 ⇒ magnified
+ve ⇒ Virtual image, Erect
-ve ⇒ Real image, Inverted

IMAGE FORMATION BY CONVEX AND CONCAVE LENSES



Convex lens

- Converging lens
- Similar to concave mirror

Concave lens

- Diverging lens
- Similar to convex mirror

Sl NO	Position of the object	Ray Diagram	Position of image	Nature of Image
1	At ∞ u = ∞		At F	Real, inverted, diminished

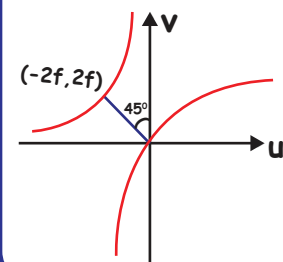
2	Beyond 2F u > 2 f		Between F and 2F	Real, inverted, diminished
3	At 2F u = 2 f		At 2F	Real, inverted, same size
4	Between F & 2F f < u < 2f		Beyond 2F	Real, inverted, enlarged
5	At F u = f		At ∞	Cannot be defined
6	within F u < f		On the side of object	Virtual, erect, enlarged

LENS FORMULA

To find v when f and u are given

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad \frac{1}{v} = \frac{1}{f} + \frac{1}{u} = \frac{u+f}{uf} \quad v = \frac{uf}{u+f}$$

U - V GRAPH

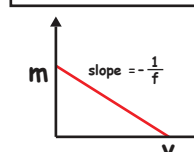


MAGNIFICATION VS V GRAPH

Concave lens

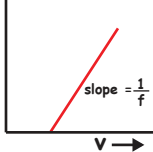
$$m = \frac{f-v}{f}$$

$$m = 1 - \frac{v}{f} = -\frac{v}{f} + 1$$

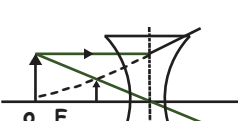


Convex lens

$$m = \frac{v}{f} + 1$$

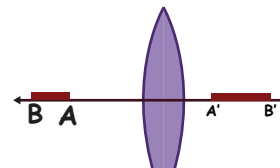


Concave lens



- Real object
- Virtual image
- Formed between F and Pole

AXIAL MAGNIFICATION



$$m_L = \frac{\text{length of image}}{\text{length of object}} = \frac{A'B'}{AB} = \frac{v_A - v_B}{u_A - u_B}$$

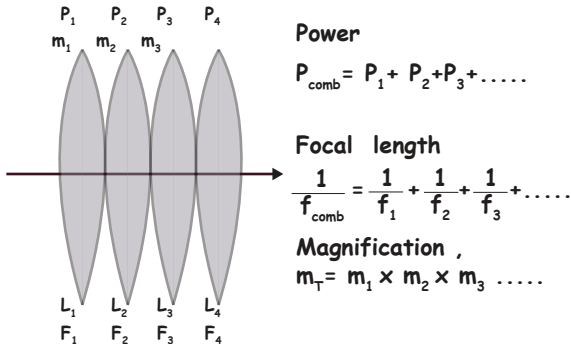
For short object
 $m_L = \frac{dv}{du} = m_L^2$



PHYSICS
WALLAH

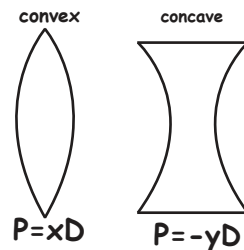
COMBINATION OF LENSES

Lenses are combined such that there is no gap between them



Note :

P_{Comb} determines whether the combination act as converging lens or diverging lens.



System acts as converging lens if total power > 0

$$P_T > 0 \\ \Rightarrow P_{\text{convex}} > P_{\text{concave}} \\ \Rightarrow f_{\text{concave}} > f_{\text{convex}}$$

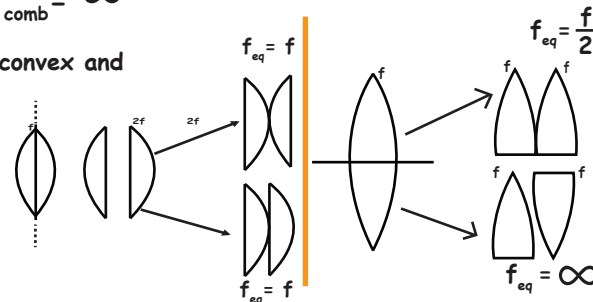
System acts as diverging lens if total power < 0

$$P_T < 0 \\ \Rightarrow P_{\text{concave}} > P_{\text{convex}} \\ \Rightarrow f_{\text{convex}} > f_{\text{concave}}$$

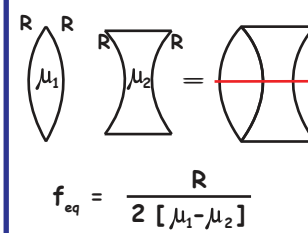
System acts as plane lens / glass if $P_{\text{comb}} = 0$

$$P_{\text{concave}} = P_{\text{convex}} = 0 \\ f_{\text{concave}} = f_{\text{convex}} = f_{\text{comb}} = \infty$$

For a combination of convex and concave lenses if $P_{\text{convex}} > P_{\text{concave}}$ $\rightarrow$ combination acts as convex lens



F_{EQ} USING LENS MAKERS FORMULA

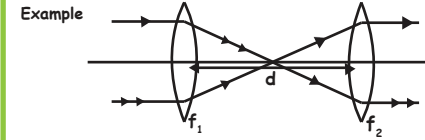


LENSES COMBINED SUCH THAT THERE IS A GAP BETWEEN THEM

Power : $P_{\text{comb}} = P_1 + P_2 - dP_1P_2$

$$\frac{1}{f_{\text{comb}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1f_2}$$

Note: Only valid if object is at ∞



If $P_{\text{comb}} = 0$

$$\frac{1}{f_{\text{comb}}} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1f_2} = 0$$

$$d = f_1 + f_2$$

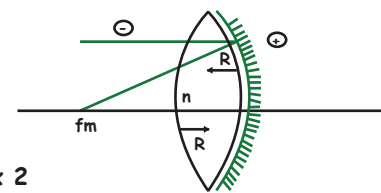
NATURE OF MIRROR IS DETERMINED BY FOCAL LENGTH

To find focal length, split the silvered lens into a lens and a mirror and apply IOS

$$u = \infty \quad I - O = S \\ v = f_m \\ R_1 = +R \quad \frac{-1}{f_m} = \frac{n-1}{R} \times 2 + \left[\frac{0+n}{+R} \right] \times 2 \\ R_2 = -R$$

[Ray passes through the surface 2 times thus multiplied by 2]

$$f_m = \frac{-R}{4n-2} = \frac{-R}{2(2n-1)}$$



When equiconvex lens is silvered (mirrored) its focal length become negative (concave mirror) with magnitude

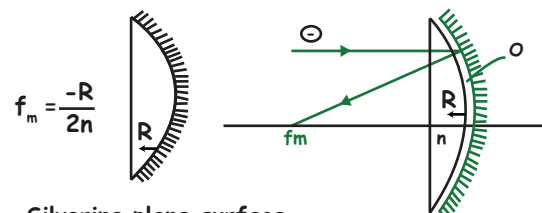
$$f_m = \frac{-R}{2(2n-1)}$$

SILVERING OF PLANO-CONVEX LENS

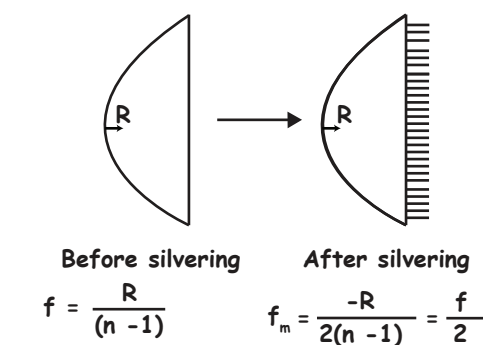
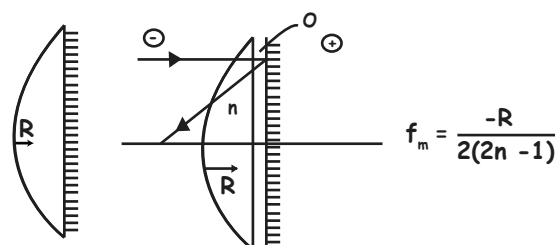
2 possibilities

- silvering curved surface
- Silvering plane surface

Silvering curved surface

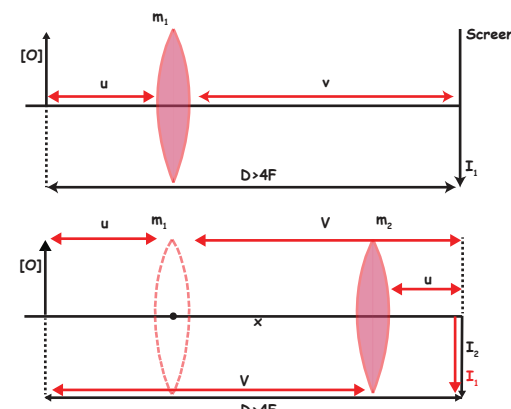


Silvering plane surface



LENS DISPLACEMENT METHOD

Distance between object & image $> 4F$



In lens Displacement method

- $D \geq 4F$
 - $F = \frac{D^2 - x^2}{4D}$ $x \rightarrow$ distance between 2 position of lens
 - $F = \frac{x}{m_1 - m_2}$
 - $m_1 m_2 = 1$ $F = \text{Focal length of lens}$
- $$[O] = \sqrt{I_1 I_2}$$

LENS MIRROR COMBINATION

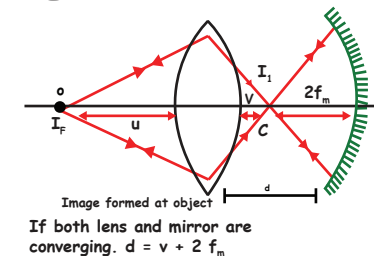


Image formed at object
If both lens and mirror are converging. $d = v + 2f_m$

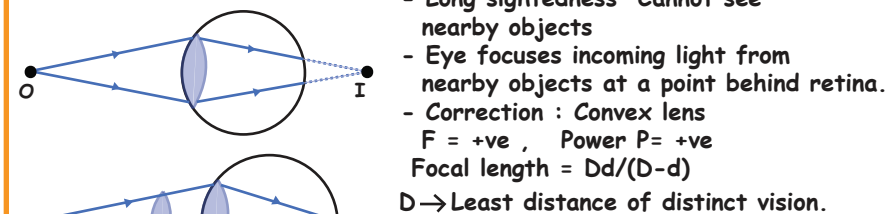
If one is converging & other is diverging $d = v - 2f_m$

HUMAN EYE

Least distance of distinct vision is 25 cm

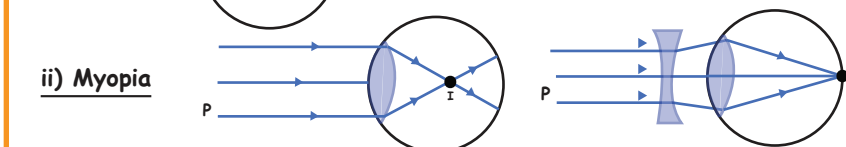
Defects of vision and their correction

i) Hypermetropia



- Long sightedness - Cannot see nearby objects
- Eye focuses incoming light from nearby objects at a point behind retina.
- Correction : Convex lens
- $F = +ve$, Power $P = +ve$
- Focal length = $Dd/(D-d)$
- $D \rightarrow$ Least distance of distinct vision.

ii) Myopia



- Short sightedness - Cannot see faraway objects
- Light from a distant object arriving at the eye lens may get converged at a point in front of the retina.
- Correction : Concave lens
- $F = -ve$, $P = -ve$, $F = -d$, $P = \frac{1}{F}$

iii) Astigmatism

eye cannot focus in horizontal and vertical planes simultaneously

- Correction : Cylindrical lens

iv) Presbyopia

eye suffers both myopia and hypermetropia

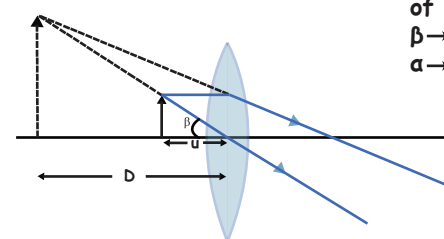
- Correction : Bifocal lens

OPTICAL INSTRUMENTS

Simple Microscope

Only one lens [convex lens]

Image is formed at least distance of distinct vision (D)
 $\beta \rightarrow$ angle subtended by image at eye.
 $\alpha \rightarrow$ angle subtended by object at eye when placed at distance D.



Case I :

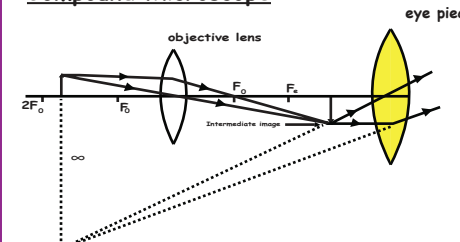
Eye under relaxed state or normal vision
Final image at infinity

Object at $f \rightarrow u = u_{\text{max}}$ $m_{\text{min}} = \frac{D}{u_{\text{max}}} = \frac{D}{f}$

Case 2 : Eye under strain

Final image at D $u = u_{\text{min}}$ $m_{\text{max}} = \frac{D}{u_{\text{min}}} = \frac{1+D}{f}$

Compound Microscope



- Has 2 lenses
- Length of microscope = Distance between lenses.
- Magnification $m = m_o \times m_e$ [m_e - same as simple microscope]

To get magnified image, object placed between F_0 & $2F_0$ of objective lens. This image is called intermediate image.

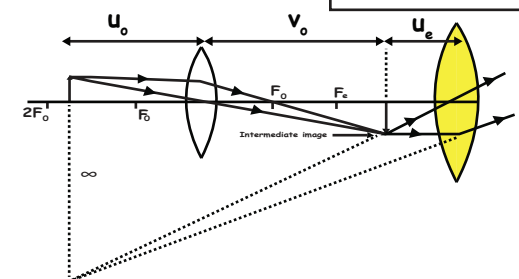
Intermediate image $\rightarrow$ real, inverted, magnified.

Intermediate image is formed within or at the focus of eyepiece.

Total magnification $m_T = -m_o \times m_e$

m_e same as simple microscope

$$m_T = -m_o \times m_e = - \left[\frac{v_o}{u_o} \times \frac{D}{u_e} \right]$$



RAY OPTICS 2

Case I :

Eye in relaxed state or final image at ∞ , $u_e = f_e$

$$m_T = m_{\min} = -\left[\frac{V_o}{u_o} \frac{D}{f_e}\right] \text{ Also } L = L_{\max} = V_o + u_{\max} = V_o + f_e$$

Case 2 : Strained eye

$$u = u_{\min} \\ L = L_{\min} = L_D = V_o + U_e = V_o + u_{\min} = V_o + \frac{Df_e}{D+f_e}$$

$$m_D = m_{\max} = \frac{-v_o}{u_o} \left[1 + \frac{D}{f_e}\right] \quad \frac{m_{\max}}{m_{\min}} = \frac{D+f_e}{D}$$

$$m_D = \left[\frac{L}{f_o}\right] \left[1 + \frac{D}{f_e}\right] \\ m_{\infty} = \left[\frac{L}{f_o} \frac{D}{f_e}\right]$$

Note :

$$L_{\infty} = V_o + U_{\max} \quad L_{\infty} = V_o + f_e$$

$$L_{\infty} = \frac{U_o f_o}{U_o - f_o} + f_e \quad L_D = V_o + U_{\min}$$

$$L_D = V_o + \frac{Df_e}{D+f_e} \text{ and } L_D = \frac{U f_o}{U_o - f_o} + \frac{Df_e}{D+f_e}$$

For microscope, eyepiece larger than objective

$$f_e \uparrow \Rightarrow m \uparrow \quad f_e \downarrow \Rightarrow m \downarrow$$

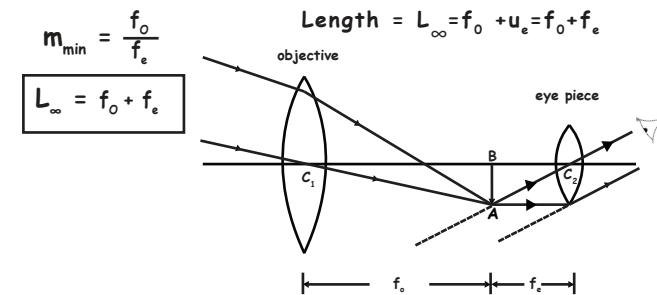
For telescope, eyepiece smaller than objective to increase magnification.

Telescope

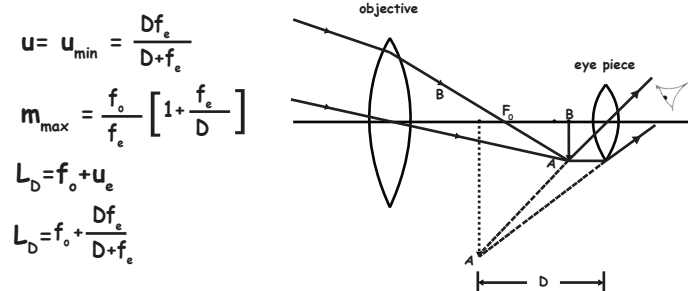
$$\text{Magnification } m = \frac{f_o}{f_e} \quad u = u_{\max} \Rightarrow m = m_{\min} \\ u = u_{\min} \Rightarrow m = m_{\max}$$

$$\text{Length of telescope } L = f_o + u_e$$

Normal adjustment /Relaxed eye/final image at ∞



Eye under strain/Final image at least distance of distinct vision.



Length of Telescope

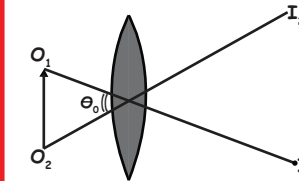
u_e can have values : f_e or $\frac{Df_e}{D+f_e}$

v_o can have values : f_o only

$$L_D = v_o + u_e$$

RESOLVING POWER

$$\text{Resolving power} = \frac{1}{\text{Resolving limit}}$$



$X = \text{Limit of resolution}$
= Resolving limit

MICROSCOPE

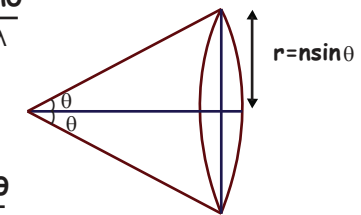
$$\text{Resolving Limit} = \frac{1.22\lambda}{a} = \frac{1.22\lambda}{2n \sin\theta}, \text{ where } a = \text{diameter of lens}$$

$$R.P. = \frac{a}{1.22\lambda} = \frac{2n \sin\theta}{1.22\lambda}$$

$$R.P. \propto \frac{1}{\lambda}$$

TELESCOPE

$$R.P. = \frac{a}{1.22\lambda} = \frac{2n \sin\theta}{1.22\lambda}$$



RAY OPTICS 2